

K_a and Percent Ionization

Separating what the acid decides from what you decide, and finding out what dilution does to each.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/ka-percent-ionization/

1 Before you open anything

Answer from what you already know. Nothing here is graded on being right — the point is to have something on paper that the demonstration can later confirm or contradict.

PREDICTION

A flask holds acetic acid, CH_3COOH , at 0.100 M. You add water until the concentration is 1.00×10^{-5} M — ten thousand times more dilute. Circle one for each:

the **percent of the acetic acid that is ionized** rises / falls / unchanged

the **pH** rises / falls / unchanged

the **K_a** rises / falls / unchanged

Write one or two sentences saying how you decided.

2 Set up one solution and read it completely

Open the demonstration. Everything on the page describes a single flask, reported several different ways.

1. Under **The acid**, press the **CH_3COOH** button. The K_a value should now read 1.8×10^{-5} , with ($\text{p}K_a$ 4.74) beside it.
2. Under **How much acid you added**, check that **Concentration** reads **0.100 M**. If it does not, use **Dilute $\times 10$** or **Concentrate $\times 10$** until it does.
3. Fill in the table from the **This solution** panel — the five tiles first, then the five rows of the species table underneath them.

What the demonstration calls it	Value
Percent ionized	
pH	
$[\text{H}_3\text{O}^+]$	
K_a – never moves	
H_3O^+ from water	
C, acid added	
$[\text{HA}]$ intact	
$[\text{A}^-]$ ionized	
$[\text{OH}^-]$	

a. Using your three recorded concentrations, evaluate $[\text{H}_3\text{O}^+][\text{A}^-] \div [\text{HA}]$. Compare the result with the K_a tile. Show the arithmetic.

b. Percent ionized is $[\text{A}^-] \div C \times 100$. Check that your own numbers give the percent the demonstration displays.

c. From the panel **Against a strong acid at the same concentration**, record the HCl column, then answer: how many times more H_3O^+ is in the HCl flask, and how many pH units apart are they?

Both flasks hold the same 0.100 mol of acid per litre. Nothing separating them is a matter of how much was added. Every value on the page is quoted at 25 °C, which is the only temperature at which $K_w = 1.0 \times 10^{-14}$ and the pH figures hold.

3 What dilution actually changes

One control moves here: concentration. K_a stays where it is, and you can watch the K_a tile to confirm that.

1. Keep CH_3COOH selected. Press **Concentrate $\times 10$** until Concentration reads **1.00 M**. Record the first row.
2. Press **Dilute $\times 10$** once and record the next row. Repeat until you reach $1.00 \times 10^{-5} \text{ M}$.
3. Fill the last two columns from your own rows, not from the screen.

Concentration	Percent ionized	pH	$[\text{H}_3\text{O}^+]$ (M)	percent $\div$ percent above	pH - pH above
1.00 M				—	—
0.100 M					
0.0100 M					
$1.00 \times 10^{-3} \text{ M}$					
$1.00 \times 10^{-4} \text{ M}$					
$1.00 \times 10^{-5} \text{ M}$					

a. In the top few rows, each ten-fold dilution multiplies the percent ionized by roughly the same factor. What is it, to two significant figures, and what is $\sqrt{10}$?

b. That factor shrinks in the bottom rows. Percent ionized there is no longer small, so one approximation behind the $\sqrt{10}$ pattern has stopped holding. Which one, and why does it fail once a large fraction of the acid has ionized?

c. By how much does the pH change per ten-fold dilution in the top rows? A strong acid diluted ten-fold changes pH by exactly 1.00 unit. Account for the difference.

d. State the rule in your own words: on dilution, what happens to percent ionization, to pH, to $[\text{H}_3\text{O}^+]$, and to K_a ?

e. Return to your part 1 prediction. Which of the three circles were borne out? For any that were not, name the specific assumption behind it.

A larger *fraction* of a smaller *amount* can still be less acid. Both columns rose at once and neither reading is wrong — check your $[\text{H}_3\text{O}^+]$ column and confirm it fell the whole way down.

4 Working backwards, and where the shortcut breaks

Exam questions rarely hand you the concentration and ask for the pH in that order. Both halves of this part run the calculation the other way.

1. With **CH₃COOH** selected, drag the **Concentration** slider until **Percent ionized** reads as close to 50% as the slider allows — near 3.6×10^{-5} M. Record the concentration and the pH.
2. Press **HF** and do the same: drag until Percent ionized is as close to 50% as you can get, near 1.35×10^{-3} M. Record concentration and pH.

Acid	Concentration at $\approx 50\%$ ionized	Percent ionized	pH there	pK _a (beside the K _a value)
CH ₃ COOH				
HF				

- a. State the relationship between pH and pK_a at 50% ionization, then prove it: start from $K_a = [\text{H}_3\text{O}^+][\text{A}^-] \div [\text{HA}]$ and impose $[\text{A}^-] = [\text{HA}]$.

- b. Reselect **CH₃COOH** and set the concentration to 1.00×10^{-5} M. Record both panels:

	Percent ionized	pH
This solution		
What the shortcut would say		

- c. The shortcut assumes x is small, so that $[\text{HA}]$ stays equal to C and $[\text{H}_3\text{O}^+] = \sqrt{(K_a C)}$. Its percent here is not merely inaccurate but impossible. Say what makes it impossible, and identify the exact step of the derivation that fails.

d. Convert both pH values in your table to $[\text{H}_3\text{O}^+]$ and find the factor by which the shortcut overstates it.

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it, set each solution up, and check — and where you disagree, find the line where it went wrong. All values are for 25 °C.

1. Formic acid, HCOOH , has $K_a = 1.8 \times 10^{-4}$. A solution is 0.0100 M. Find $[\text{H}_3\text{O}^+]$, pH and percent ionization by solving the equilibrium exactly — the quadratic, not the approximation. Then compute what the “x is small” approximation would have given, and state the percentage by which it misses $[\text{H}_3\text{O}^+]$.

2. Hypochlorous acid, HClO , has $K_a = 3.0 \times 10^{-8}$. A solution is 0.100 M. Find $[\text{H}_3\text{O}^+]$, pH and percent ionization. You may assume x is small here — but finish by showing the check that justifies the assumption.

3. Two flasks each contain 0.100 mol of acid per litre: one HF ($K_a = 6.8 \times 10^{-4}$), one HCl. Find the pH of each and the ratio of their hydronium concentrations. Then explain, in terms of what K_a measures, why identical amounts of acid give pH values more than a unit apart.

4. At 1.00×10^{-5} M, the “x is small” approximation reports acetic acid as 134.2% ionized. Without calculating the exact answer, explain which assumption produced a number above 100%, why every genuine percent ionization must lie below it, and whether the true value is larger or smaller than 134.2%. State also what happens to K_a between 0.100 M and 1.00×10^{-5} M.

6 On your own

Nothing here is collected. The demonstration holds more than this worksheet used.

- Select **HClO** (pK_a 7.52) and dilute it all the way to 1.00×10^{-5} M. It never comes near 50% ionized. Using $pH = pK_a$ at 50%, and the fact that the pH curve leans on the pH 7 line without crossing it, work out why that ceiling exists for any acid with pK_a above 7.
 - Leave the presets behind and drag the K_a slider itself. Set it to its maximum, 1.0×10^{-3} , at 0.100 M and see how ionized the strongest acid the slider allows actually gets. Then dilute until it reaches 50% and note the concentration.
 - Select **HCN**, press **Watch it dilute**, and keep your eye on the **H₃O⁺ from water** tile. Find roughly where water starts supplying more than half of the hydronium, and decide what still makes it reasonable to call that flask an acid solution.
 - Read the **Honest limits** panel. Name one thing the model leaves out that a real 1 M flask would show, and say which direction it would push the numbers.
-